VTU INTERNSHIP PROJECT

PROJECT TITLE:

CUSTOMER CHURN ANALYSIS

Predicting Customer Retention using Data Analytics

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INTRODUCTION

Customer churn refers to the phenomenon where customers discontinue using a company's product or service over a period of time. It is a critical concern for businesses in sectors such as telecommunications, banking, and e-commerce, as high churn rates directly impact revenue and growth. Analyzing customer behavior and identifying patterns that lead to churn can help organizations take proactive measures to retain customers. This project focuses on leveraging data analytics techniques using PySpark and Power BI to understand customer usage trends, detect potential churn indicators, and provide meaningful insights that support better decision-making and customer retention strategies.

PROBLEM STATEMENT

Businesses face challenges in retaining customers due to a lack of clear insights into customer behavior. Large volumes of customer data remain underutilized, making it difficult to identify potential churn. Without accurate analysis, companies cannot take timely actions to improve retention. Therefore, a data-driven approach is needed to analyze patterns and predict customer churn effectively.

OBJECTIVES

- Analyze customer behavior using historical data
- Perform feature engineering to improve prediction
- Use PySpark for large-scale data processing
- Visualize churn insights using Power BI
- Help businesses improve retention strategies

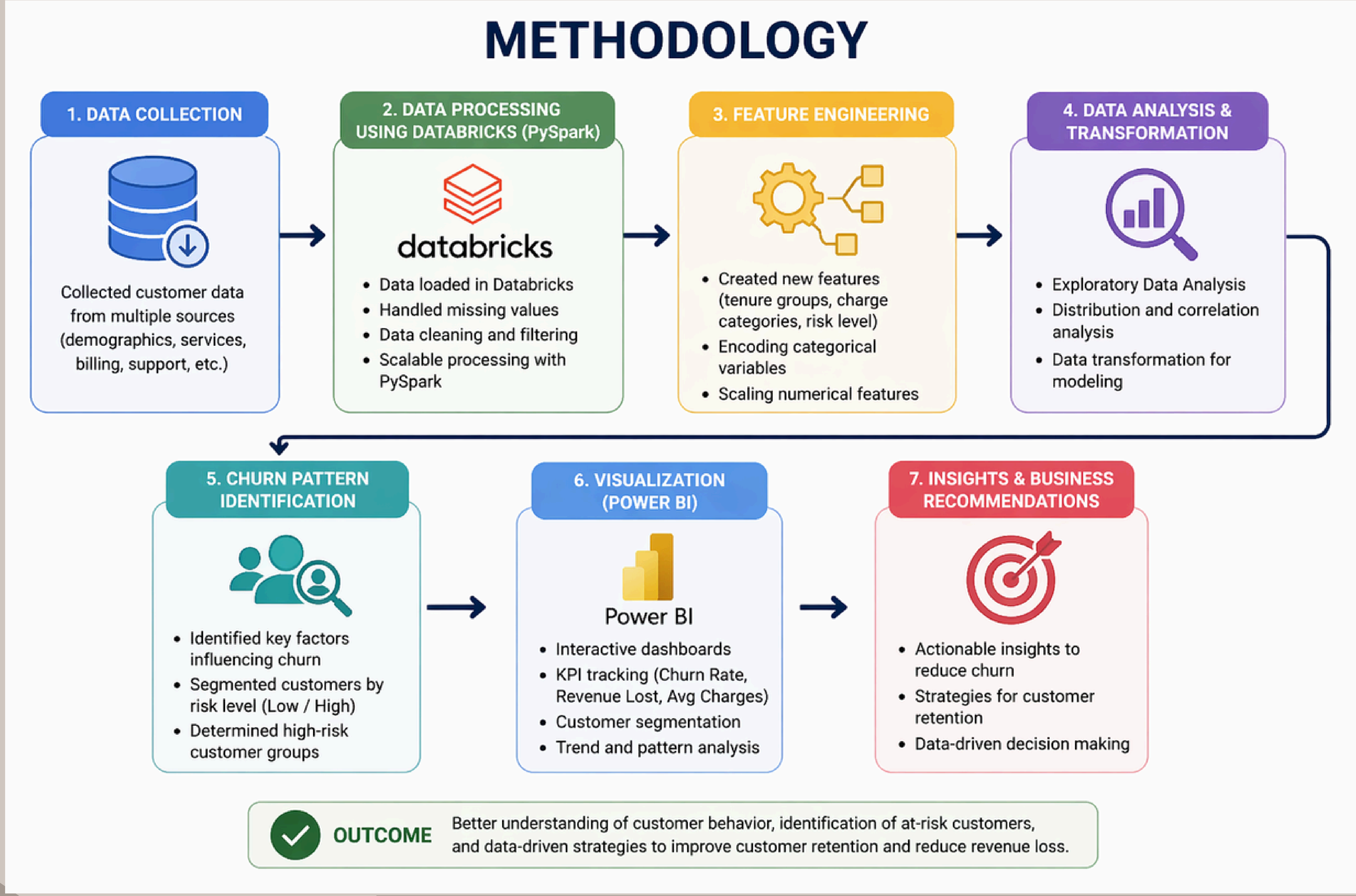
TOOLS & TECHNOLOGIES

databricks

MySQL™

Power BI

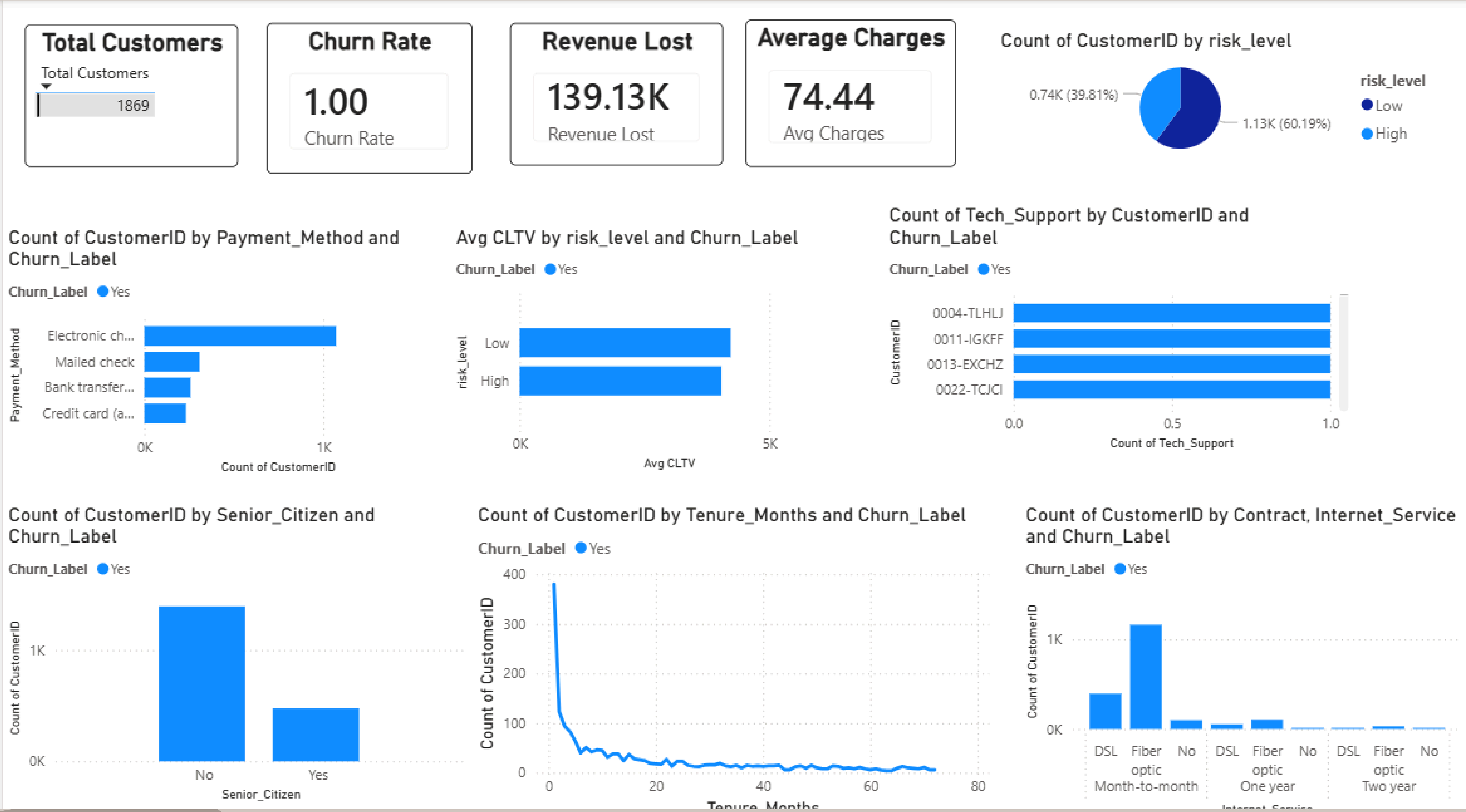
METHODOLOGY



RESULTS

- Successfully analyzed customer's churn patterns using Power BI dashboard
- Identified that month-to-month contract customers have the highest churn rate
- Observed that customers with low tenure are more likely to leave early
- Found that electronic check payment method shows higher churn compared to others
- Detected that customers without tech support are more prone to churn
- Identified high-risk customers using tenure-based segmentation
- Calculated revenue loss due to churn, highlighting business impact
- CLTV analysis showed that low-risk customers contribute higher long-term value
- Dashboard enabled data-driven insights for improving customer retention strategies

RESULTS



Total Customers
Total Customers
1869
Churn Rate
1.00
Churn Rate
Revenue Lost
139.13K
Revenue Lost
Average Charges
74.44
Avg Charges
Count of CustomerID by risk_level
0.74K (39.81%)
1.13K (60.19%)
risk_level
Low
High
Count of CustomerID by Payment_Method and Churn_Label
Churn_Label Yes
Payment_Method
Electronic ch...
Mailed check
Bank transfer...
Credit card (a...
0K
1K
Count of CustomerID
Avg CLTV by risk_level and Churn_Label
Churn_Label Yes
risk_level
Low
High
0K
5K
Avg CLTV
Count of Tech_Support by CustomerID and Churn_Label
Churn_Label Yes
CustomerID
0004-TLHLJ
0011-IGKFF
0013-EXCHZ
0022-TCJCI
0.0
0.5
1.0
Count of Tech_Support
Count of CustomerID by Senior_Citizen and Churn_Label
Churn_Label Yes
Count of CustomerID
1K
0K
No
Yes
Senior_Citizen
Count of CustomerID by Tenure_Months and Churn_Label
Churn_Label Yes
Count of CustomerID
400
300
200
100
0
0
20
40
60
80
Tenure_Months
Count of CustomerID by Contract, Internet_Service and Churn_Label
Churn_Label Yes
Count of CustomerID
1K
0K
DSL
Fiber optic
No
Month-to-month
DSL
Fiber optic
No
One year
DSL
Fiber optic
No
Two year

CONCLUSION

This project successfully analyzes customer churn by utilizing PySpark for efficient data processing and feature engineering, along with Power BI for interactive visualization. By identifying key factors influencing churn, such as contract type, tenure, and service usage, the analysis provides valuable insights into customer behavior. These insights enable businesses to take proactive measures to improve customer retention and reduce revenue loss. Overall, the project demonstrates the importance of data-driven decision-making in enhancing business performance and customer satisfaction.

THANK YOU